Terek Johnson

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Education

University of California, Riverside — Riverside, CA
Master of Science in Computer Science, GPA: 3.90 — *Sep 2026 – Expected June 2027*

University of California, Riverside — Riverside, CA
Bachelor of Science in Computer Science, GPA: 3.51 — *Sep 2024 – June 2026*

Coursework: Software Engineering, Computer Architecture, Embedded Systems, Robotics, Design & Synthesis of Digital Systems, Physics (Electricity & Magnetism), Logic Design

Experience

Software Engineering Intern — June 2026 – Present
NASA Jet Propulsion Laboratory — *Pasadena, CA*

- Developing a portable Kaitai Struct (KSY) library to read, parse, and serialize Deep Space Network binary data across 27 formats
- Wrote Python scripts that parse binary files into JSON and serialize structured data back to binary
- Built generative AI "skills" that draft KSY specifications from legacy format documentation, reducing manual spec-authoring effort

Transfer Student Peer Mentor — Sep 2025 – June 2026
Marlan and Rosemary Bourns College of Engineering, University of California, Riverside — *Riverside, CA*

- Directed social media and marketing campaigns, growing non-follower engagement by 8%
- Advised hundreds of transfer and prospective students on academics, research, and career pathways
- Supervised engineering study spaces and represented the college at outreach events and campus tours

Undergraduate Research Assistant — June 2024 – Sep 2024
Collaborative Intelligent Systems Lab, University of California, Riverside — *Riverside, CA*

- Collected and synchronized a multi-modal LiDAR and camera dataset with a team of eight, using instrumented vehicles and city sensors to support autonomous-driving machine learning (ML) pipelines
- Evaluated data quality and debugged sensor drivers to validate the dataset for ML training and benchmarking
- Presented findings at the 2024 Bourns College of Engineering Summer Research Symposium

Projects

AHOY! (Artificial Horizon from Orientation & Yaw) | *FPGA, Python, SPI, UDP* — June 2026

- Designed an FPGA-based (IceSugar-Pro) artificial horizon display modeled on CubeSat avionics, using a complementary filter and CORDIC angle conversion to compute real-time orientation from IMU data over SPI
- Built a Raspberry Pi Pico 2W telemetry link streaming live orientation data over WiFi (UDP/lwIP) to a Python ground station

Single-Axis Solar Tracker & Wireless Telemetry (Senior Capstone) | *Arduino, nRF, GPS* — Feb 2026

- Engineered a single-axis solar tracker using lux sensors, an Arduino Nano, and an H-bridge-driven DC motor to orient a solar panel toward peak sunlight for a statewide collegiate solar boat competition
- Built a wireless transceiver system using nRF antennas, integrating a GPS module, an IMU, a DC power sensor, and lux sensors to transmit live telemetry from the lake to a shore-side station

Technical Skills

Languages: Python, C++, C, C#, Java, JavaScript, HTML/CSS
Frameworks & Libraries: React, TailwindCSS, SFML, Kaitai Struct
Developer Tools: Git, GitHub, Docker, MongoDB, Figma, JSON, CI/CD, Agile/Scrum, JIRA
Embedded Systems: FPGA, ASIC, Raspberry Pi Pico, Arduino, Sensing, Control
Hardware & Lab: Multimeter, Soldering
Protocols: SPI, I2C, UART, PWM, UDP
AI/ML: Claude Code, Windsurf, GitHub Copilot, OpenAI API, Agent Skills, Computer Vision, Natural Language Processing